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Toward Carbon Neutrality in Vietnam: Analyzing the Role of Renewable Energy Consumption and Economic Growth Using Wavelet Coherence and Quantile-on-Quantile Approaches

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ABSTRACT

The rapid economic growth and rising energy demand in Vietnam have resulted in increased CO₂ emissions, mainly due to a significant reliance on fossil fuels. Specifically, Vietnam has substantial potential for green energy sources, which are vital to the current model of economic growth. Therefore, this study investigates the dynamic connection among carbon emissions, renewable energy usage, and economic growth in Vietnam during 1985–2023 by applying wavelet coherence and quantile-on-quantile regression (QQR) approaches. The results indicate that, over the sample period, significant co-movements and lead-lag patterns among the employed variables are identified mainly in localized short-run episodes, with more limited evidence at medium frequencies. The findings reveal stronger and more recurrent coherence between economic growth and carbon emissions than between renewable energy consumption and carbon emissions; these relationships are mostly in phase within significant regions, with changing lead-lag dynamics across sub-periods. Moreover, the QQR results show that economic growth is associated with higher CO₂ emissions at high quantiles of CO₂ emissions, whereas the utilization of green energy has a conditional effect: it is positively correlated with carbon emissions at some medium quantiles of carbon emissions; however, green energy consumption reduces carbon emissions at high quantiles of renewable energy use and CO₂ emissions. To achieve net-zero greenhouse gas emissions by 2050, the empirical evidence indicates that accelerating the transition to renewable energy is vital for sustainable growth and improved environmental quality.

1 | Introduction

The escalating threat of global warming and environmental degradation highlights the connection between economic development and environmental pollution, with carbon emissions (CO₂ emissions) identified as the major cause of these problems. Vietnam and other developing countries must achieve a balance between sustaining rapid economic expansion and reducing

carbon emissions. The Vietnamese economy has experienced substantial growth over the past three decades, with GDP per capita rising from \$238 in 1985 to over \$4323 in 2023 [1]. However, rapid industrialization and urbanization have also contributed to rising CO₂ emissions, particularly in an economy that still relies heavily on traditional energy sources. As a result, CO₂ emissions increased from 0.3 tons per capita in 1985 to 3.3 tons per capita in 2023 [2].

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Furthermore, the depletion of traditional fuels and their harmful effects on the environment make the development of sustainable and environmentally friendly energy substitutes increasingly important worldwide. The Vietnamese government is increasingly focused on reducing CO₂ emissions and has implemented a number of policies to achieve this goal. During the 26th United Nations Climate Change Conference of the Parties (COP26) in 2021, the Vietnamese government pledged to achieve net zero emissions. However, Vietnam's utilization of fossil fuels has grown significantly since reform and opening up, resulting in a noticeable deterioration in the country's environmental quality. In Vietnam, the consumption of fossil fuels directly contributes to rising CO₂ emissions, with coal being the largest source of emissions by fuel type or industrial sector, accounting for 57% of total carbon emissions [1]. Several renewable energy policies have been introduced, which have promoted the continued development of renewable energy consumption. The Vietnamese government, through Power Development Plan VIII, aims to increase the share of clean energy consumption in the total electricity consumption, with the proportion of wind, solar, hydro, gas, and coal being 27%, 34%, 7%, 13%, and 19%, respectively, by 2050 [3]. Nevertheless, Vietnam's share of electricity production remains mainly dependent on fossil fuels, with coal accounting for 46.87% of the total, followed by hydropower (29.26%), natural gas (9.52%), solar (9.3%), wind (4.11%), and oil and bioenergy (0.93%), respectively, in 2023 [4]. Therefore, the key to guaranteeing Vietnam's sustainable growth in energy, economy, and environment is to increase the utilization of green energy and decrease the consumption of fossil fuels.

The relationship among the utilization of renewable energy, pollution, and economic growth remains inadequately analyzed [5]. Research examining the potential of green energy as a feasible strategy for decreasing emissions has been conspicuously lacking in the pursuit of green growth economics. Previous studies on the relationship between economic growth and emissions in Vietnam and other countries have primarily employed traditional time-series or panel methodologies.

The novelty of this study lies in integrating wavelet coherence and quantile-on-quantile regression (QQR) within a single country-specific framework for Vietnam. Existing Vietnam-based studies either employ wavelet analysis alone or rely on conventional time-series and regression approaches. This study jointly examines both time–frequency dependence and distributional asymmetry in the CO₂ emissions–economic growth–renewable energy consumption nexus over 1985–2023. The contribution of this study is threefold. First, it determines whether the relationships concentrate in the short, medium, or long run. Second, it reveals that the effects of economic growth and renewable energy on CO₂ emissions are nonlinear, asymmetric, and regime-dependent. Third, by combining both methods, the study shows that the emissions-reducing role of renewable energy in Vietnam is conditional rather than uniform, becoming more visible only when renewable energy use and emission intensity are sufficiently high.

The methodological justification for integrating wavelet coherence and QQR is grounded in the complementary analytical strengths of the two approaches and is particularly relevant to the Vietnamese context. Wavelet coherence allows the study to identify co-movements that change over time between CO₂

emissions, economic growth, and renewable energy consumption across different frequency bands, while also revealing temporal lead and lag relationships that conventional regressions in the time domain may not fully detect. However, it does not show whether these relationships differ across emission regimes. By contrast, QQR captures distributional heterogeneity by examining how different quantiles of economic growth or renewable energy consumption are associated with different quantiles of CO₂ emissions, although it does not identify the specific time–frequency domains in which such dependence occurs. This methodological integration is therefore useful for Vietnam, where rapid economic transformation, increasing electricity demand, dependence on fossil fuels, and the relatively recent expansion of modern renewable energy have created complex and evolving emissions dynamics. In such a transitional context, estimates based on a single average relationship may obscure substantial heterogeneity across time horizons and emission regimes. By jointly applying wavelet coherence and QQR, this study identifies not only whether economic growth and renewable energy consumption are associated with CO₂ emissions, but also when these associations occur, at which frequency horizons the associations are most evident, and under which quantile-specific conditions the effects of economic growth and renewable energy consumption change. This integrated framework offers a more comprehensive and policy relevant understanding of the nexus among economic growth, renewable energy consumption, and CO₂ emissions in Vietnam than approaches that rely only on mean estimators or a single method in the time and frequency domains.

The structure of this study is as follows: The relevant literature is analyzed in Section 2, the data and methodologies are discussed in Section 3, the empirical results and discussion are provided in Section 4, and the study is concluded with policy implications in Section 5.

2 | Literature Review

Many studies have investigated the connections between CO₂ emissions, economic development, and the utilization of renewable energy through varying methodologies; however, a consensus has not been reached. Numerous studies have used wavelet analysis to examine the connections between green energy utilization, economic growth, and CO₂ emissions. The findings indicate that the utilization of green energy reduces carbon emissions, whereas the use of nonrenewable energy and economic expansion generally increase emissions [6–10]. The findings suggest that promoting investments in renewable energy and gradually phasing out fossil fuels are crucial for achieving environmental sustainability and development objectives [6, 7, 10]. Additionally, while most renewable technologies help reduce emissions, some cutting-edge technologies, including wind and solar, may have little or even adverse effects on CO₂ emissions in EU countries [11].

Abban et al. [7] employed wavelet coherence and Toda–Yamamoto causality to analyze the relationship among economic growth, emission of carbon dioxide, and clean energy resources in African oil-producing countries. The findings demonstrated a positive correlation between economic growth and emissions, while a negative correlation was found with

renewable energy. Utilizing wavelet coherence techniques in Algeria, Chekouri et al. [12] found strong relationships between emission of carbon dioxide, energy utilization, and economic development across medium- and long-term horizons. Moreover, Soyulu et al. [9] identified a positive correlation between GDP growth and carbon dioxide emissions, as well as energy consumption, across all frequencies in China from 1965 to 2019, utilizing the wavelet approach. Alqaralleh and Hatemi-J [13] demonstrated that energy utilization has a substantial impact on economic growth, highlighting the importance of renewable energy sources in eight countries. Adebayo et al. [14] employed Morlet wavelet analysis from 1980 to 2019 to investigate the relationship between CO₂ emissions and economic development, renewable energy utilization, trade openness, and technological innovation in Portugal. The results show that the utilization of green energy decreases CO₂ emissions in both the medium and long term in Portugal. Bilgili et al. [15] applied the continuous wavelet analysis to assess the impact of renewable energy on the US economy at 1–3 year and 3–8 year frequencies, spanning from January 1989 to November 2016, and significant effects of solar, wind, geothermal, wood, biofuels, and waste were shown for both short- and long-term cycles. Le [16] investigated the dynamic interrelationship between green energy utilization, GDP, and CO₂ emissions, concentrating on the high-frequency interval from 1985 to 2001 and the low-frequency interval from 2011 to 2019.

Moreover, many studies have examined the relationships among green energy, economic development, and CO₂ emissions using quantile regression techniques. Many studies have shown that using renewable energy reduces emissions, and the impacts are stronger at higher emission levels [17–20]. Heal [21] and Forsberg [22] noted that renewable energy might not lead to a reduction in emissions due to its intermittent output and insufficient storage technology. Consequently, during periods of high electricity demand, a backup power source is necessary, typically supplied by fossil fuels. Shahbaz and Patel [18] and Jeon [19] found an inverted U-shaped relationship between GDP and emissions, as described by the environmental Kuznets curve. Saidi and Omri [23] demonstrated that green energy has led to an increase in emissions of carbon dioxide in the Netherlands and South Korea. Shahbaz and Patel [18] underscored the imperative to advocate for the utilization of renewable energy, enhancement of energy efficiency, and the implementation of suitable environmental laws to facilitate sustainable growth. The results indicated that energy intensity and utilization of fossil fuels both increase CO₂ emissions, while environmental taxes and clean energy production significantly decrease carbon emissions, in particular at higher quantiles. Altinoz and Dogan [24] employed quantile regressions to examine the effects of the supply of natural resources and utilization of renewable energy on carbon dioxide emissions in a panel of 82 nations. The findings indicated that the utilization of renewable energy resources reduces emissions, with a stronger impact at higher quantiles.

Additionally, Xu et al. [8], Huo et al. [25], and AlNemer et al. [6] found a strong relationship between CO₂ emissions, GDP, and the use of fossil fuels across a number of quantiles and periods. The relationship between renewable energy resources and emissions is intricate. Several studies indicate a negative correlation [26], whereas others report varying results depending on the time

period, region, and economic context [7, 27]. Moreover, other studies have emphasized the need to recognize factors including trade openness [9], economic development [26], and globalization [25].

Recent empirical studies indicate that the relationship among economic growth, energy use, and carbon emissions depends strongly on a country's specific conditions, the estimation method applied, and the time horizon analyzed. Using wavelet-based techniques, Khochiani and Nademi [28] showed that the relationships among economic growth, energy use, and CO₂ emissions differ substantially across the United States, China, and India, especially across short-, medium-, and long-run horizons. Likewise, using QQR, Yu et al. [17] demonstrated that the environmental effect of solar-based renewable energy varies across countries and across the joint distribution of renewable energy use and CO₂ emissions. These studies suggest that conventional mean-based models may mask important nonlinearities, asymmetries, and regime-dependent effects. This evidence strongly supports the use of a combined wavelet coherence and quantile-on-quantile approach for Vietnam, where rapid structural transformation, continued reliance on fossil fuels, and a still-developing renewable energy sector may lead to relationships that change over time and vary across different conditions.

Moreover, recent studies have extended the renewable energy–environment nexus by incorporating institutional quality, technology progress, policy uncertainty, human capital, and broader sustainability indicators. Pata [29] found that economic growth, financial development, and urbanization increase environmental degradation, while renewable energy consumption was not at a desirable level to reduce CO₂ emissions in Turkey. Empirical findings indicate that the environmental benefits of renewable energy depend on institutional quality, technology, human capital, and policy uncertainty [30, 31]. Meanwhile, Pata et al. [32] show that income growth promotes renewable energy deployment before generating carbon-emission reduction effects, highlighting the sequential link between growth, energy transition, and environmental improvement. Pata and Pata [33] examined the effects of energy policy uncertainty, climate policy uncertainty, and energy transition on renewable energy mineral production in China, focusing on cobalt, lithium, graphite, and rare earth elements. Using multivariate QQR and Fourier-based quantile causality tests, the study found that energy policy uncertainty negatively affects only graphite production, while climate policy uncertainty promotes lithium, graphite, and rare earth production across several quantiles. The findings further show that energy transition consistently increases the production of all examined renewable energy minerals, highlighting the importance of policy stability and clean-energy investment for securing China's low-carbon mineral supply chain. Moreover, Sethi et al. [34] showed that positive shocks to green growth, renewable energy, and political stability contribute to ecological footprint mitigation in BRICS+ countries, while information and communication technology (ICT) has an asymmetric effect and is associated with an increase in ecological footprint. These studies suggest that the environmental benefits of renewable energy are not automatic but depend on institutional quality, technological capacity, human capital, policy stability, and country-specific transition conditions. These findings also support the use of nonlinear and context-sensitive methods when examining the

relationships among economic growth, renewable energy, policy uncertainty, and environmental outcomes.

Table 1 provides a summary of several previous research studies that employed different approaches to examine the connections between CO₂ emissions and economic growth and between CO₂ emissions and the use of green energy sources.

3 | Data and Methodology

3.1 | Data

This study uses annual data on GDP per capita (in current US dollars) from the World Bank's World Development Indicators database; annual CO₂ emissions per capita (tons per person); and renewable energy consumption per capita (kWh-equivalent),

TABLE 1 | Summary of previous studies using various methodologies.

Authors	Time	Countries	Methodologies	Results
Charfeddine and Kahia [35]	1980–2015	24 countries of MENA	Panel vector autoregressive	REC → CO ₂
Bilgili et al. [36]	1977–2010	17 OECD countries	FMOLS DOLS	REC → CO ₂
Pao and Tsai [37]	1980–2007	BRIC countries	Panel co-integration	REC → CO ₂
Zoundi [38]	1980–2012	25 African countries	ARDL	REC → CO ₂ CO ₂ → GDP
Al-Mulali and Ozturk [39]	1990–2012	27 countries	Panel FMOLS VECM	REC ↔ CO ₂
Dong et al. [40]	1985–2016	BRICS countries	Panel VECM Granger causality	REC ↔ CO ₂
Lu [41]	1990–2012	24 Asian countries	Panel co-integration Granger causality	REC ↔ CO ₂
Saidi and Mbarek [42]	1990–2012	9 developed countries	VECM Granger causality	REC ↔ CO ₂
Menyah and Wolde-Rufael [43]	1960–2007	US	Vector autoregressive model Granger causality	CO ₂ → REC
Shafiei and Salim [44]	1980–2011	OECD countries	Panel co-integration Granger causality	CO ₂ → REC GDP → CO ₂
Apergis et al. [45]	1984–2007	19 countries	Panel cointegration Granger causality tests	REC ≠ CO ₂
Bento and Moutinho [46]	1960–2011	Italy	ARDL	REC ≠ CO ₂
Lin and Moubarak [47]	1977–2011	China	ARDL bounds test Granger causality	REC ≠ CO ₂
Dong et al. [48]	1965–2016	China	VECM	GDP → CO ₂
Wang et al. [49]	1995–2012	30 China's provinces	Panel co-integration Granger causality	GDP → CO ₂
Al-Mulali et al. [50]	1990–2013	23 European countries	Panel VECM	GDP ↔ CO ₂ REC → CO ₂
Riti et al. [51]	1970–2015	China	ARDL, VECM	GDP ↔ CO ₂
Rafindadi [52]	1971–2011	Nigeria	ARDL VECM Granger causality test	GDP ↔ CO ₂
Solarin et al. [53]	1965–2013	China, India	ARDL VECM	GDP ↔ CO ₂
Shahbaz et al. [54]	1975–2011	Indonesia	VECM	GDP ↔ CO ₂
Han et al. [55]	1996–2014	China	Granger causality	GDP ≠ CO ₂

Note: “→,” “↔,” and “≠” indicate unidirectional, bidirectional, and no causality relationships, respectively. ARDL, DOLS, FMOLS, and VECM are autoregressive distributed lag model, dynamic ordinary least squares, fully modified ordinary least square, and vector error correction model, respectively. CO₂, GDP, and REC represent CO₂ emissions, economic growth, and renewable energy consumption, respectively.

TABLE 2 | Descriptive statistics of variables in natural logarithm form.

Statistic	LnCO ₂	LnGDP	LnREC
Mean	−0.027757	6.647702	6.491487
Median	0.101699	6.388977	6.507637
Maximum	1.307138	8.371786	8.155963
Minimum	−1.304832	4.586059	4.179648
Std. Dev.	0.830461	1.144869	1.081500
Skewness	0.003917	−0.027918	−0.521693
Kurtosis	1.661944	1.793715	2.678100
Jarque-Bera	2.909489	2.369643	1.937443

Note: LnCO₂, LnGDP, and LnREC are logarithms of CO₂ emissions, economic growth, and renewable energy consumption, respectively.

which are sourced from the Our World in Data database. The study focuses on the years 1985–2023. The descriptive data are shown in Table 2.

In this study, the series for GDP per capita (GDP), CO₂ emissions per capita (CO₂), and renewable energy utilization per capita (REC) are all represented using natural logarithms. The log transformation has two primary functions: (i) it stabilizes connections for regression analysis by linearizing exponential growth patterns and (ii) it makes it possible to interpret regression coefficients as elasticity or semi-elasticity. In wavelet analysis, using logged series assists in stabilizing variance. At extremely low frequencies outside of the visible range, the study focuses on coherence significance rather than raw correlation in order to identify patterns.

The Jarque–Bera test statistics are below the critical value at the 5% significance level, indicating that the null hypothesis of normality cannot be rejected and that the variables are approximately normally distributed. Furthermore, the mean and median of LnGDP are closely aligned with those of LnREC. Stationarity is not a significant concern in the QQR analysis due to the nonparametric character of the methodology; however, this study conducted the Augmented Dickey–Fuller (ADF) unit root test on each series for analysis. The results showed that all of the variables are stationary at their first differences, as presented in Table 3.

TABLE 3 | Unit root test results.

Variables	ADF
LnCO ₂	0.117 (1)
LnGDP	−0.363 (1)
LnREC	−1.658 (1)
ΔLnCO ₂	−5.482 ^a (1)
ΔLnGDP	−5.431 ^a (1)
ΔLnREC	−4.504 ^a (1)

Note: Δ denotes the first difference of values.

^aRepresents significance at the 1%.

3.2 | Methodology

Economic growth affects CO₂ emissions mainly through expansion in production, transport, industrial output, and overall energy demand. In an economy where fossil fuels still account for a large share of the energy mix, higher GDP is therefore expected to increase CO₂ emissions. By contrast, greater renewable energy use can reduce emissions when it substitutes for coal, oil, and gas in electricity generation and final energy consumption. Since CO₂ emissions are thought to be the main cause of climate change, global warming, and environmental damage, the studies are concentrated on reducing them [56, 57].

From a theoretical perspective, the general relationship is specified as in Equation (1):

$$\text{LnCO}_{2t} = \alpha_0 + \alpha_1 \text{LnGDP}_t + \alpha_2 \text{LnREC}_t + \varepsilon_t, \quad (1)$$

where LnCO_{2t} is the natural logarithm of carbon dioxide emissions per capita at time *t*, LnGDP_{*t*} is the natural logarithm of gross domestic product per capita, LnREC_{*t*} is the natural logarithm of renewable energy consumption per capita, α₀ is the intercept term, α₁ and α₂ are the elasticity coefficients, and ε_{*t*} is the error term, α₁ is expected to be positive and α₂ is expected to be negative. Because wavelet coherence and QQR are implemented in a bivariate setting, the empirical analysis operationalizes this relationship through two separate linkages: (i) the growth–emissions nexus, represented by the pair (LnCO₂ and LnGDP) and (ii) the renewable-energy–emissions nexus, represented by the pair (LnCO₂ and LnREC). This strategy allows the study to examine each channel separately across time, frequency, and quantiles while keeping the interpretation transparent. Therefore, the empirical specifications are written as:

$$\text{LnCO}_{2t} = \beta_0 + \beta_1 \text{LnGDP}_t + u_t, \quad (1a)$$

$$\text{LnCO}_{2t} = \mu_0 + \mu_1 \text{LnREC}_t + v_t, \quad (1b)$$

where *u_t* and *v_t* are disturbance terms. The parameters β₀ and μ₀ denote intercept terms, while β₁ and μ₁ are slope coefficients capturing the elasticity of carbon emissions with respect to the corresponding explanatory variable.

Figure 1 summarizes the empirical procedure. After data collection and log transformation, the study first conducts descriptive statistics and ADF unit root tests, then applies wavelet coherence and QQR, and finally integrates the findings to derive policy implications.

3.2.1 | Wavelet Coherence Analysis

Wavelet coherence analysis has important advantages over traditional correlation analysis because it allows the relationship between two time series to vary simultaneously over time and across frequencies. The wavelet coherence methodology, formulated by Goupillaud et al. [58], is employed to analyze periodic events in time series, particularly in instances where abrupt frequency changes occur over time. This technique measures the degree and amount of co-movement between two time series variables (*a*, *b*) within the time–frequency domain.

The squared wavelet coherence between two series, *a* and *b*, is defined as in Equation (2):

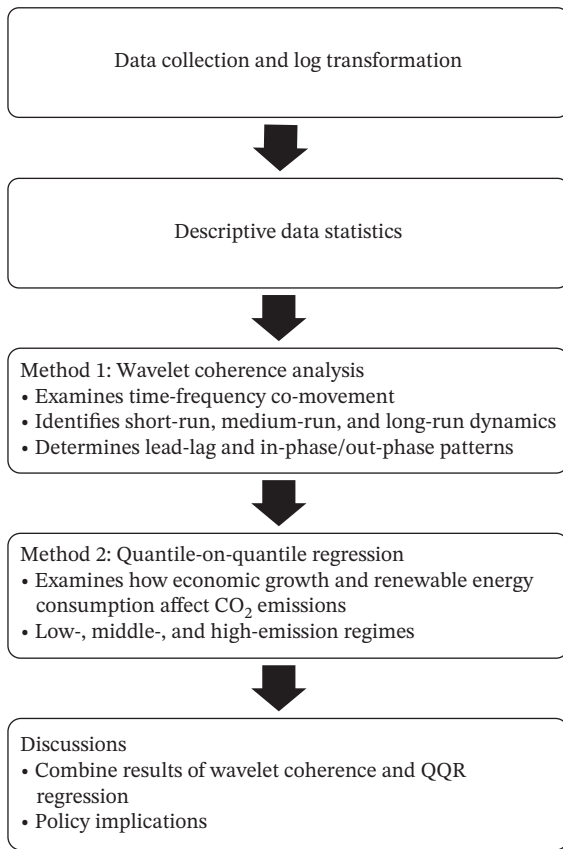


FIGURE 1 | Flowchart of the analytical procedure from data collection to empirical results and policy implications.

$$R_{ab}(u, s) = \frac{\zeta[W(a, b; u, s)]}{\sqrt{\zeta[W(a; u, s)] \cdot \zeta[W(b; u, s)]}}, \quad (2)$$

$$R_{ab}^2(u, s) = R_{ab}(u, s) \cdot R_{ab}^*(u, s)$$

where u denotes the time location and s denotes the wavelet scale, which is interpreted as the frequency horizon. $R_{ab}^2(u, s)$ is the measure of squared wavelet coherence between series a and b at time position u and scale s , and R_{ab} , which is the measure of wavelet coherence between series a and b . $\zeta(\cdot)$ denotes the smoothing operator. When the squared wavelet coherence is near 1, it suggests a strong correlation between the time series at a certain scale, which is represented by the red color and bounded by a black line. Conversely, when the squared wavelet coherence is approximately 0, it indicates a lack of association between the time series, shown in blue.

$W(a, b)$ is the cross-wavelet that is defined by:

$$W_{ab}(u, s) = W_a(u, s) \cdot W_b^*(u, s). \quad (3)$$

In Equation (3), $W_b^*(u, s)$ is the complex conjugate of the wavelet transform of the series b . $W_{ab}(u, s)$ is the cross-wavelet transform between series a and b , $W_a(u, s)$ and $W_b(u, s)$ are the continuous wavelet transforms of series a and b , respectively. The smoothing operator with the necessary time–frequency resolution is denoted by ζ [59].

Nevertheless, the squared wavelet coherence by itself is unable to differentiate between in-phase and out-of-phase co-movements. The phase difference between the series is analyzed using the

methodology established by Torrence and Compo [59] to identify any delays in the movements of the series. The phase difference at a specified time and frequency is defined as in Equation (4):

$$\Psi_{a,b}(u, s) = \tan^{-1} \left(\frac{\Lambda[W_{a,b}(u, s)]}{\Gamma[W_{a,b}(u, s)]} \right), \quad (4)$$

where Γ and Λ represent the real and imaginary components, respectively, of the cross-wavelet transform $W_{a,b}$. The graphic layout of the wavelet coherence display features the horizontal axis denoting the time dimension, whereas the vertical axis represents frequency. The color red signifies a strong association between the series, while the color blue denotes a weaker relationship. The cold regions outside the relevant zone indicate the frequency and duration, revealing no correlation between the series. Rightward arrows indicate in-phase co-movement, whereas leftward arrows indicate out-of-phase co-movement. Upward arrows imply that the first variable leads the second, while downward arrows imply that the second variable leads the first. Accordingly, for the pairs (LnCO₂ and LnGDP) and (LnCO₂ and LnREC), the interpretation of lead and lag always follows the ordering in which the variables are presented in each figure.

Wavelet coherence is suitable for this study because it shows how co-movements between variables evolve across time and frequency, while phase differences indicate lead-lag dynamics. This is useful in the Vietnamese context, where shifts in the energy mix may alter the GDP–CO₂ and REC–CO₂ relationships across short-, medium-, and long-run horizons.

3.2.2 | QQR

Building upon the well-established framework of the traditional quantile regression model, QQR is a unique technique for bivariate analysis first introduced by Sim and Zhou [60]. The QQR integrates conventional quantile regression with nonparametric estimation, offering more comprehensive insights than traditional estimation techniques while addressing their limitations. The QQR technique examines a thorough relationship between the lower and upper quantiles of the data series, incorporates potential for structural breaks, and can function effectively in an asymmetric setting.

The QQR method represents the quantile of the dependent variable (LnCO₂) as a function of the quantile of the independent variable (X). In this study, the q -th conditional quantile of logged CO₂ emissions is expressed as a function of the explanatory variable X_t , where X_t may denote either LnGDP _{t} or LnREC _{t} . The quantile indexes $q \in (0, 1)$ and $p \in (0, 1)$ refer to the quantiles of the dependent variable and the explanatory variable, respectively. The component u_t^q indicates error term of the quantiles where the conditional q -th is proposed equal to zero. The QQR model is specified as in Equation (5):

$$Q_{\text{LnCO}_2}^q(X_t) = \theta^q(X_t) + u_t^q. \quad (5)$$

A first-order Taylor expansion is applied to approximate the unknown link function $\theta^q(X_t)$ around the p -th quantile of the explanatory variable X , where $\theta_0(q, p)$ denotes the local intercept and $\theta_1(q, p)$ denotes the local slope coefficient measuring the

effect of the p -th quantile of X_t on the q -th quantile of LnCO_2 . X^p denotes the p -th quantile of the explanatory variable X_t .

$$\begin{aligned} \theta^q(X_t) &\approx \theta_0(q, p) + \theta_1(q, p)(X_t - X^p) \\ \text{with} \\ \theta_0(q, p) &= \theta^q(X^p) \\ \theta_1(q, p) &= \frac{\partial \theta^q(X^p)}{\partial X} \Big|_{X=X^p} \end{aligned} \quad (6)$$

Substituting the first-order Taylor expansion in Equation (6) into the QQR model yields the following:

$$Q_{\text{LnCO}_2}^q(X_t) \approx \theta_0(q, p) + \theta_1(q, p)(X_t - X^p) + u_t^q. \quad (7)$$

Equation (7) represents the local linear form of the QQR model. The coefficients $\theta_0(q, p)$ and $\theta_1(q, p)$ vary across the quantiles of the dependent and independent variables, thereby capturing the dependence structure between the q -th quantile of LnCO_2 and the p -th quantile of X .

The empirical analysis is organized around two separate variable pairs: CO_2 emissions–economic growth nexus and CO_2 emissions–renewable energy consumption nexus. Accordingly, the estimated relationships should be interpreted as conditional pairwise dependence rather than as fully structural causal effects. This empirical design is consistent with the objective of the study, which is to uncover time–frequency comovement and distributional asymmetry in the two core channels of Vietnam’s growth–renewable energy–emissions nexus. Because the QQR technique is a bivariate regression strategy that precludes the inclusion of additional control variables in the model, the study includes only one independent variable and one dependent variable in the functional form of the QQR model. Despite its bivariate nature, the QQR approach remains useful for the present study because it captures asymmetric and state-dependent effects across the joint distribution of the variables. The approach evaluates the asymmetric impacts of the independent variable’s quantiles on the dependent variable’s quantiles, focusing on both the lower and upper quantiles within the data distribution, and thereby provides more detailed and robust results compared to traditional methodologies [61]. Because the baseline specification does not directly include additional macroeconomic and energy-related controls, the QQR results should be taken as evidence of asymmetric pairwise dependence rather than conclusive causal effects.

The Equation (8) is minimized to derive the estimated value of Equation (7):

$$\min_{\theta_0, \theta_1} \sum_{i=1}^n \rho_q \left[\text{LnCO}_{2i} - \theta_0(q, p) - \theta_1(q, p)(X_i - X^p) \right] \times K \left(\frac{G_n(X_i) - p}{h} \right), \quad (8)$$

where $G_n(X_i)$ denotes the empirical cumulative distribution function of the explanatory variable X_i . The Gaussian kernel $K(\cdot)$ is used to minimally weight the observation using a normal distribution based on bandwidth h . Following Sim and Zhou [60], this study adopted a bandwidth of 5%. The Gaussian kernel is symmetrical about zero and gives closer observations higher weights.

QQR complements wavelet coherence by examining how the effect of either LnGDP or LnREC on LnCO_2 varies across the

joint distribution of the variables. Rather than imposing a single average coefficient, it allows the relationship to differ across low- and high-emission regimes, thereby identifying heterogeneous effects that may not be visible in mean-based estimates.

The use of a quantile-based framework is also supported by the broader environmental econometrics literature. Environmental datasets often exhibit heterogeneity and nonconstant variance, so mean-based estimators may obscure relationships that are strong only in specific parts of the conditional distribution. Quantile methods are particularly useful when upper pollution levels are policy-relevant because they allow the marginal effect of an explanatory variable to vary across emission regimes rather than imposing a single average coefficient [62]. In this study, this supports the use of QQR to examine whether different quantiles of economic growth and renewable energy exert heterogeneous effects on different quantiles of CO_2 emissions in Vietnam, thereby revealing how low-, medium-, and high-growth or renewable-energy states interact with low-, medium-, and high-emission states.

4 | Empirical Results and Discussion

Descriptive patterns suggest that the relationships among GDP, CO_2 emissions, and renewable energy consumption in Vietnam are not stable over the full period. GDP and CO_2 emissions rose markedly over 1985–2023, whereas renewable energy expanded mainly in the later years; accordingly, the analysis next applies wavelet coherence and QQR to examine whether these linkages differ across time, frequency, and quantiles.

To ensure a clear link between the data and the findings, the results are presented in the same order as the methodology. First, wavelet coherence is used to identify time–frequency co-movements and lead–lag patterns between LnCO_2 and LnGDP and between LnCO_2 and LnREC . Second, QQR is used to examine whether the effects of LnGDP and LnREC on LnCO_2 differ across low-, medium-, and high-emission regimes. The results are then compared with evidence from other countries to highlight the Vietnam-specific contribution of the study.

4.1 | Wavelet Coherence

Figures 2 and 3 illustrate the wavelet coherence results for the pairs (CO_2 emissions and economic growth) and (CO_2 emissions and renewable energy utilization). Red areas bounded by black contours indicate statistically significant co-movement, while the arrows show phase differences and lead–lag relationships.

The wavelet coherence between CO_2 emissions and GDP is shown in Figure 2. Analysis of the heat map indicates a significant correlation between CO_2 emissions and GDP in the short term (0–4 years) from 1988 to 1994. Since the arrows mainly point up and to the right, CO_2 emissions lead GDP and the two variables move in phase. From 2008 to 2015, a significant correlation between CO_2 emissions and GDP was observed, with arrows directed down-right, indicating a positive co-movement in which GDP leads CO_2 emissions. After 2015, the phase pattern becomes more mixed across high-frequency bands, implying that the short-run relationship between GDP and CO_2 emissions is time-varying. Overall, the strongest evidence for the GDP– CO_2

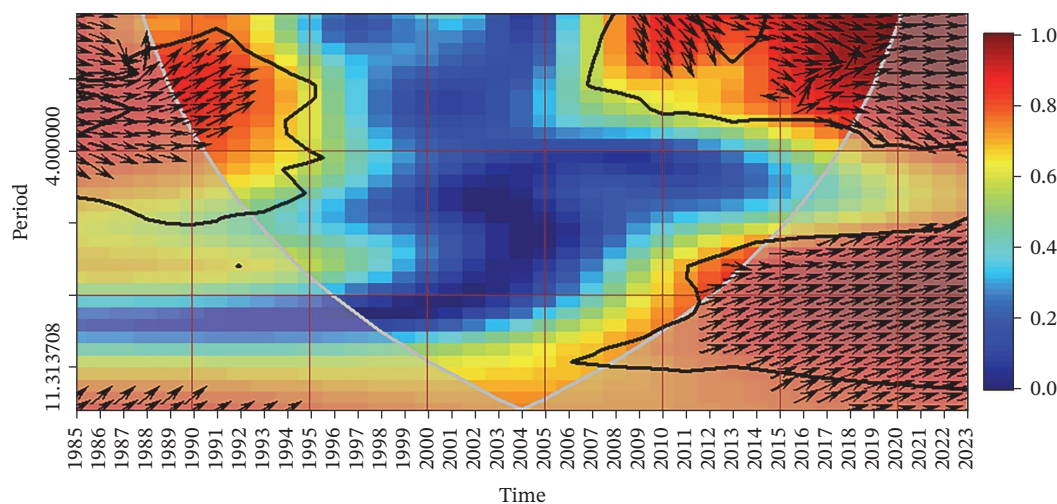


FIGURE 2 | Wavelet coherence representing CO₂ emissions and GDP.

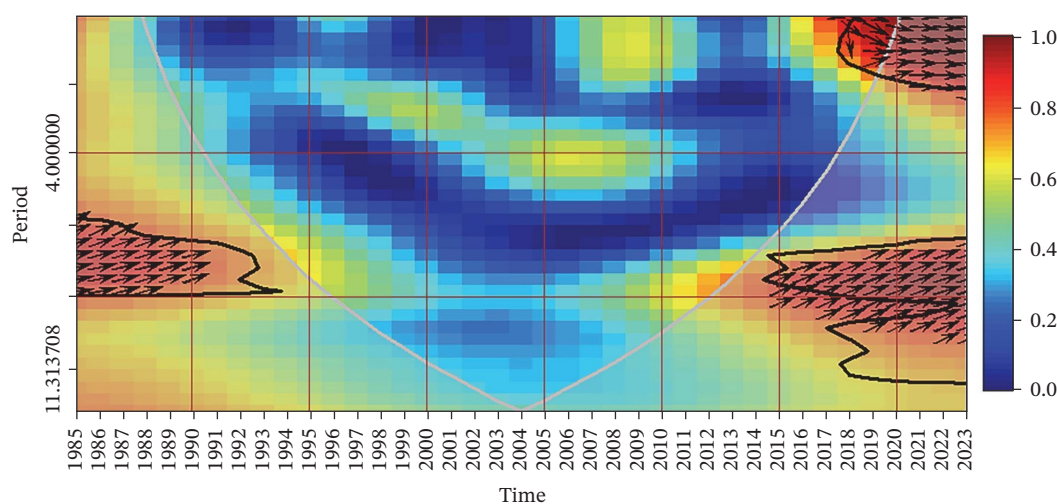


FIGURE 3 | Wavelet coherence representing CO₂ emissions and renewable energy consumption.

emissions nexus is concentrated in the short run, while coherence is weaker over much of the medium- and long-run ranges. Similarly, Ahmed et al. [63], in a study covering 162 nations, confirmed a similar correlation between CO₂ emissions and economic development, and coherence was low in the long term and medium term. This implies that there is no direct relationship between annual changes in emissions and annual changes in GDP.

Figure 3 illustrates the wavelet coherence between carbon emissions and green energy consumption, revealing low coherence before 2015 at most frequencies because the percentage of renewable energy was small, primarily derived from hydropower prior to 2008 [64], which indicates minimal dynamic interaction with emissions. During that period, emissions were primarily influenced by the growing supply of fossil fuels, such as coal and oil, rather than changes in the renewable energy composition. Over the long term (scale 8–12), the results did not show significant coherence. The structural link between renewable energy sources and overall emissions may not have had sufficient time to establish a clear long-term trend, or this lack of clarity could be attributed to the limitations of our sample length in detecting

very low-frequency relationships. However, the arrows were primarily heading downward and rightward in the short term at scale 2 from 2017 to 2019, indicating that CO₂ emissions and renewable energy were positively co-moving, with renewable energy consumption leading the way. Thus, the findings suggest that, in some short-run range after 2015, renewable energy is positively associated with emissions in Vietnam. Although the proportion of renewable energy per capita increased significantly from 13.9% to 23.1% from 2010 to 2017, the proportion of fossil energy in the total energy remained high, accounting for 76.9% in 2017, causing CO₂ emissions and renewable energy to tend to co-move [2]. Moreover, while the manufacturing process for wind turbines and solar photovoltaic panels adds to CO₂ emissions, more recent and sophisticated renewable energy sources, including wind and solar, are orders of magnitude less polluting than other traditional fossil fuel energies like coal, natural gas, and oil. Therefore, it is possible to justify the positive impact on these coefficients. Additionally, it is probable that Vietnam has not yet attained the point at which CO₂ emissions may be reduced by renewable energy. Similar results are found by Yu et al. [17] and Chen et al. [65] for solar energy in France and different

regions of China (solar energy impacted CO₂ emissions positively) and by Heal [21] and Forsberg [22].

After 2015, a significant correlation between CO₂ emissions and green energy is observed in both the short run (scale 2–4) and the long run (scale 8–12). Although some areas of higher coherence also appear after 2015 in both the short run and the long run band, much of this evidence is limited in scope and partly located near the cone of influence; therefore, it should not be treated as robust long-run evidence. It can be stated that the connection between renewable energy and CO₂ emissions appears weak and insufficient to explain long-term trends, consistent with the findings of Apergis et al. [45]. The implementation of green energy sources failed to achieve a reduction in emissions. This outcome aligns with the findings of Menyah and Wolde-Rufael [43] and Al-Mulali et al. [66].

Lastly, the wavelet results suggest that the CO₂ emissions–GDP nexus in Vietnam is stronger and more visible in specific short-run, sub-periods, whereas the CO₂ emissions–REC nexus is weaker over much of the sample and appears only in localized episodes. These findings indicate time-varying co-movement and changing lead-lag dynamics rather than a single stable relationship over the period 1985–2023.

These patterns are consistent with Vietnam’s evolving energy structure. The strong short-run coherence between GDP and CO₂ emissions in the earlier sub-period reflects output expansion associated with energy-intensive production and fossil fuel use. By contrast, the later positive short-run comovement between renewable energy and CO₂ emissions suggests that renewable energy appears to have been added alongside fossil fuel expansion rather than fully substituting for it.

4.2 | QQR

Figure 4 shows that the connection between GDP and CO₂ emissions varies significantly across CO₂ emissions quantiles. At lower quantiles, GDP has only a limited impact and, in some regions, is associated with lower emissions, reflecting a period when the economy is less environmentally intensive. However, negative coefficients are observed mainly in some middle-quantile regions, particularly around the range of 0.4–0.6, rather than uniformly across all quantiles. This may suggest that, in those

quantile regions, economic growth was accompanied by improvements in technology and environmental management that helped moderate emissions. However, a more pronounced positive association is observed for the medium-to-higher quantiles of GDP (0.6–0.9) and the high quantiles of CO₂ emissions (0.7–0.9). These findings suggest that the environmental impact of GDP is nonlinear and asymmetric across quantiles, which may be consistent with heterogeneous growth–emissions dynamics in Vietnam, although they do not constitute formal confirmation of the environmental Kuznets curve. This pattern is broadly consistent with studies reporting nonlinear or state-dependent growth–emissions relationships, such as Tang and Tan [67], Shahbaz and Patel [18], Jeon [19], Minh et al. [68], and Lean and Smyth [69], although it differs from Al-Mulali et al. [66], who did not find evidence supporting the environmental Kuznets curve in Vietnam.

Figure 5 shows that the impact of green energy consumption on CO₂ emissions is mixed across quantiles. There is only a weak correlation between the lower-middle to higher quantiles of green energy consumption and the lower quantiles of CO₂ emissions (0.1–0.3). This suggests that the utilization of renewable energy does not have a significant impact on reducing carbon emissions. However, the beta coefficient is positive in the middle quantiles of CO₂ emissions (0.4–0.6), particularly at the low and middle quantiles of green energy use, indicating that green energy is insufficient to replace fossil fuels and may potentially be linked to an increase in emissions.

This positive association should not be interpreted as evidence that renewable energy is inherently highly polluting. Rather, it reflects Vietnam’s transitional energy structure. During the sample period, renewable energy was not always deployed as a direct substitute for fossil fuels; in several phases, it expanded alongside rising total energy demand and continued fossil-fuel generation. Therefore, renewable energy consumption and CO₂ emissions may increase together when the scale effect of economic activity and electricity demand dominates the substitution effect of renewable energy. This interpretation is consistent with Vietnam’s power-sector context, where coal has remained the largest electricity source in recent years [70]. Although Vietnam’s Eighth National Power Development Plan sets out a long-term transformation toward a cleaner power structure, the expansion of renewable energy has not yet fully replaced electricity generation

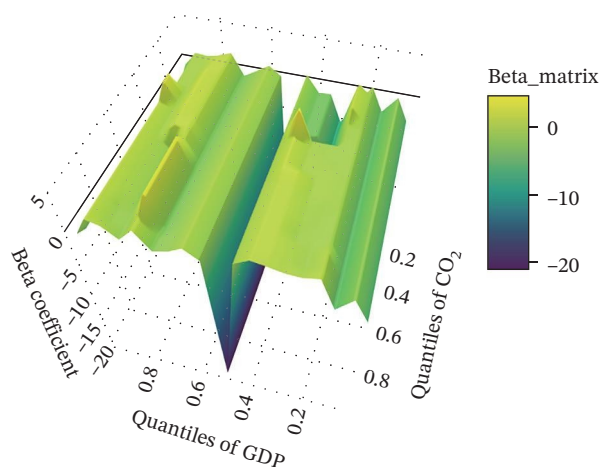


FIGURE 4 | QQR estimates of the CO₂ emissions–GDP.

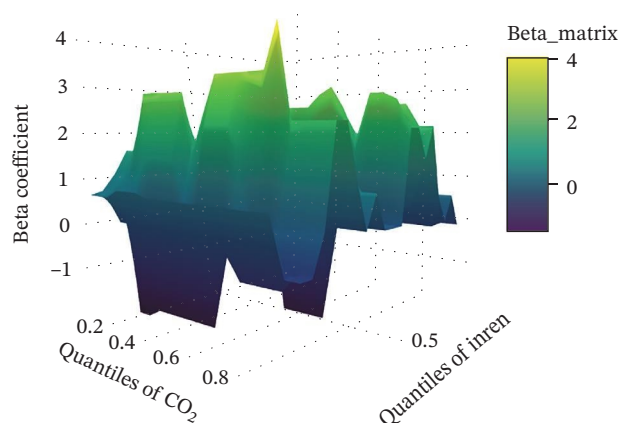


FIGURE 5 | QQR estimates of the CO₂ emissions–renewable energy consumption.

reliant on fossil fuels [3]. This explanation is also consistent with the argument that intermittency and insufficient storage may limit the immediate emissions-reducing effect of renewable energy [21, 22].

In addition, the positive CO₂ emissions–renewable energy consumption relationship may reflect integration constraints. When renewable energy penetration remains relatively limited, the intermittency of solar and wind generation may require continued reliance on fossil fuels to provide backup capacity. At the same time, grid congestion, storage constraints, and delays in transmission infrastructure investment may weaken the extent to which renewable energy can effectively replace electricity generation based on coal and gas. Consequently, renewable energy deployment may not lead to immediate emissions reductions unless it reaches sufficient scale and is accompanied by grid modernization, adequate storage capacity, and flexible power-system operation.

By contrast, there is a negative association for the combination of the high quantiles of emissions (0.7–0.9) and the high quantiles of renewable energy consumption (0.7–0.9), suggesting that utilization of green energy improves environmental quality by reducing CO₂ emissions during high-emission scenarios.

4.3 | Discussion

The results indicate that the CO₂ emissions–GDP–renewable energy consumption nexus in Vietnam is heterogeneous across time horizons and emission regimes. In particular, the GDP–CO₂ linkage is more evident in localized short-run episodes, whereas the emissions-reducing role of renewable energy becomes clearer only at higher quantiles of renewable energy use and CO₂ emissions.

This pattern is broadly consistent with wavelet-based evidence from other countries, although the sign, strength, and timing of these relationships differ across contexts. In many cases, wavelet coherence points to a predominantly positive association between GDP and CO₂ emissions. For example, Khochiani and Nademi [28] report positive coherence between GDP and CO₂ emissions across all frequencies in the United States; Adebayo et al. [14] document positive co-movement over both short- and long-run horizons in Portugal; and Soylu et al. [9] identify positive relationships across all frequencies in China. However, this relationship is not universal. Irfan et al. [71] find negative coherence between GDP and CO₂ emissions in the United Kingdom, with GDP leading CO₂ emissions. The literature also points to marked time-scale heterogeneity. For instance, Khalfouli et al. [72] identify cyclical relationships in the G7 countries at the 1–2 year scale.

Compared with other studies, Vietnam appears less uniform over time. While studies for Portugal, China, and African oil-producing countries generally report clearer and more persistent GDP–CO₂ emissions or renewable energy consumption–CO₂ emissions linkages, the Vietnamese case shows a more localized short-run CO₂ emissions–GDP relationship and a weaker overall CO₂ emissions–renewable energy consumption linkage. This suggests that Vietnam's growth–energy–CO₂ emissions nexus remains strongly shaped by its transitional energy structure, characterized by rapid structural change, persistent fossil fuel dependence, and relatively late expansion of renewable energy.

The CO₂ emissions–renewable energy consumption results are likewise consistent with the broader literature showing that the environmental effect of renewable energy is not uniform across countries or technologies. Wang et al. [11] further indicate that some renewable technologies in European countries may have limited or even adverse effects on CO₂ emissions. Broader quantile-regression evidence also suggests that the emissions-reducing effect of renewable energy becomes stronger at higher quantiles in the global panel of 192 countries [20]. Similarly, Sharif et al. [73] report significant negative associations between renewable energy consumption and CO₂ emissions in seven of the 10 most polluted countries, whereas positive associations are observed in India, Russia, and Indonesia. In Vietnam, renewable energy does not reduce emissions uniformly over the full sample period; instead, its beneficial role becomes more visible only under high emission and high renewable energy consumption conditions.

Compared with conventional mean-based estimators, the combined wavelet coherence and quantile-on-quantile framework is better suited to the Vietnamese case because it captures both time–frequency heterogeneity and distributional asymmetry that may not be captured by full-sample average estimates. Under such conditions, full-sample mean estimators may conceal important heterogeneity. Wavelet coherence identifies when and at which frequencies the variables co-move, while QQR reveals how the effects of GDP and renewable energy differ across low-, medium-, and high-emission regimes. However, the framework also has limitations: wavelet coherence does not establish structural causality, and QQR remains essentially bivariate.

5 | Conclusion and Policy Implications

This study examines the CO₂ emissions–GDP–renewable energy consumption nexus in Vietnam over the period 1985–2023 using wavelet coherence and QQR. The results show that both the growth effect and the renewable energy effect are heterogeneous across time horizons and emission regimes.

The findings indicate that the GDP–CO₂ emissions linkage is stronger and more visible during localized short-run episodes, whereas the renewable energy effect is weaker over much of the sample and becomes more clearly emissions-reducing only at higher quantiles of renewable energy use and CO₂ emissions. Overall, renewable energy does not automatically reduce emissions in Vietnam; its environmental benefit becomes stronger only after deployment reaches a sufficient scale, while economic growth exerts a more pronounced positive effect on CO₂ emissions at higher GDP and emission quantiles.

These findings provide several policy implications specific to Vietnam. First, because economic growth is more strongly associated with CO₂ emissions at higher quantiles, Vietnam should reduce the carbon intensity of growth, particularly in electricity generation, cement, steel, transport, and energy-intensive manufacturing. This requires stricter energy-efficiency standards, more effective emissions monitoring, cleaner production requirements, and the integration of emission-control measures into industrial and infrastructure planning.

Second, since renewable energy reduces CO₂ emissions mainly when both renewable energy consumption and CO₂ emissions

are at high levels, Vietnam should expand renewable energy to a scale sufficient to replace electricity generation from coal and gas, rather than merely adding renewable sources to total energy supply. This requires substantial investment in grid infrastructure, transmission capacity, energy storage, and flexible operation of the power system. Without these supporting conditions, renewable energy may continue to expand alongside fossil fuels, which may explain the positive renewable energy consumption–CO₂ emissions association observed in some quantiles.

Third, because the wavelet results show that these relationships vary over time, Vietnam should adopt flexible energy and climate policies that can be adjusted in response to changes in electricity demand, the share of renewable energy in the power system, prices of fossil fuels, and grid conditions.

This study has several limitations. First, the sample consists of annual observations for 1985–2023, which is appropriate for long-horizon analysis but limits the number of observations available for distributional estimation. Second, wavelet coherence is informative about comovement and phase differences, but it does not by itself establish structural causality. Third, the QQR framework is essentially bivariate and therefore does not explicitly incorporate additional control variables in the main specification. Consequently, the empirical framework does not explicitly account for broader macroeconomic, structural, demographic, and energy-market conditions that may also influence CO₂ emissions.

Future research can extend this study in several directions. First, potential omitted variable bias should be addressed by using multivariate empirical frameworks, such as multivariate quantile regression, time-varying parameter models, ARDL, FMOLS, or DOLS. These approaches could incorporate additional determinants of CO₂ emissions, including trade openness, urbanization, financial development, industrial structure, energy prices, and foreign direct investment. By including these factors, future studies could examine whether the pairwise relationships between GDP, renewable energy consumption, and CO₂ emissions remain robust after controlling for broader macroeconomic, structural, and energy-market conditions. Second, future studies should examine renewable energy by specific source, such as hydropower, solar, wind, and biomass, since CO₂ emissions effects may differ depending on technological characteristics, grid integration, storage capacity, and capacity to replace fossil fuel consumption.

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Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

The data are available upon request from the author.

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